

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **SPCL_GTM-04 (AN)** **Date: 14.01.2025**
Time: 3.00Hrs. **JEE MAIN MODEL** **Max. Marks: 300**

PHYSICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

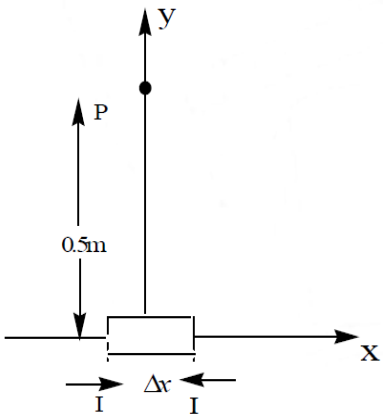
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The specific heat at constant pressure of real gas obeying $PV^2 = RT$ equation is:

A) R B) $C_v + R$ C) $\frac{R}{3} + C_V$ D) $C_V + \frac{R}{2V}$

2. An element $\overrightarrow{\Delta l} = \Delta x\hat{i}$ is placed at the origin and carries a large current $I = 10A$. The magnetic field on the y-axis at a distance of 0.5m from the elements Δx of 1 cm length is:

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A) 10×10^{-8}T B) 4×10^{-8}T C) 12×10^{-8}T D) 8×10^{-8}T

3. A sample contains mixture of helium and oxygen gas. The ratio of root mean square speed of helium and oxygen in the sample, is:

A) $\frac{1}{2\sqrt{2}}$ B) $\frac{1}{32}$ C) $\frac{2\sqrt{2}}{1}$ D) $\frac{32}{1}$

4. σ is the uniform surface charge density of thin spherical shell of radius R. The electric field at any point on the surface of the spherical shell is:

A) $\sigma/\in_0 R$ B) $\sigma/4\in_0$ C) $\sigma/\in_0$ D) $\sigma/2\in_0$

5. Which of the following phenomena does not explain by wave nature of light.

a) reflection b) diffraction c) photoelectric effect

d) interference e) Polarization

Choose the most appropriate answer from the options given below:

A) b, d only B) c only C) e only D) a, c only

6. To find the spring constant (k) of a spring experimentally, a student commits 2% positive error in the measurement of time and 1% negative error in measurement of mass.

The percentage error in determining value of 'k' is:

A) 4% B) 5% C) 3% D) 1%

7. While measuring diameter of wire using screw gauge the following readings were noted. Main scale reading is 1 mm and circular scale reading is equal to 42 divisions.

Pitch of screw gauge is 1 mm and it has 100 divisions on circular scale.

The diameter of the wire is $\frac{x}{50} mm$. The value of x is:

A) 21 B) 71 C) 42 D) 142

8. To project a body of mass m from earth's surface to infinity, the required kinetic energy is (assume, the radius of earth is R_E, g = acceleration due to gravity on the surface of earth).

A) $4mgR_E$ B) $2mgR_E$ C) mgR_E D) $1/2\ mgR_E$

9. A train starting from rest accelerates uniformly up to a speed of $80 km/h$ for time t, then it moves with a constant speed for time $3t$. The average speed of the train for this duration of journey will be (in km/h):

A) 70 B) 30 C) 40 D) 80

10. Given below are two statements:

Statement I: In an LCR series circuit, Current is maximum at resonance.

Statement II: Current in a purely resistive circuit can never be less than that in a series LCR circuit when connected to same voltage source.

In the light of the above statements, choose the correct from the options given

A) Statement I is true but Statement II is false

B) Statement I is false but Statement II is true

C) Both Statement I and Statement II are false

D) Both Statement I and Statement II are true

11. A light string passing over a smooth light pulley connects two blocks of masses m_1 and m_2 vertically (where $m_2 > m_1$). If the acceleration of the system is $\frac{g}{\sqrt{2}}$, then the ratio of the masses $\frac{m_1}{m_2}$ is:

A) $\frac{1+\sqrt{5}}{\sqrt{5}-1}$ B) $\frac{1+\sqrt{5}}{\sqrt{2}-1}$ C) $\frac{\sqrt{3}+1}{\sqrt{2}-1}$ D) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

12. In photoelectric experiment energy of 2.48 eV irradiates a photo sensitive material. The stopping potential was measured to be 0.5 V. Work function of the photo sensitive material is:

A) 2.48 eV B) 0.5 eV C) 1.68 eV D) 1.98 eV

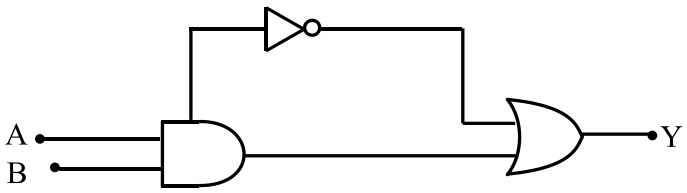
13. A bullet of mass 50 g is fired with a speed 100 m/s on a plywood and emerges with 40 m/s. The percentage loss of kinetic energy is:

A) 44% B) 32% C) 84% D) 16%

14. Four particles A, B, C, D of mass $\frac{m}{2}, m, 2m, 4m,$ have same momentum, respectively. The particle with maximum kinetic energy is:

A) C B) D C) B D) A

15. The correct truth table for the following logic circuit is:



A)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

B)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

C)

A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1

D)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

16. A small ball of mass m and density ρ is dropped in a viscous liquid of density ρ_0. After sometime, the ball falls with constant velocity. The viscous force on the ball is:

A) $mg\left(\frac{\rho_0}{\rho}-1\right)$ B) $mg\left(1+\frac{\rho}{\rho_0}\right)$ C) $mg\left(1-\frac{\rho_0}{\rho}\right)$ D) $mg\left(1-\rho\rho_0\right)$

17. Match List I with List II

	List – I		**List II**
a)	Torque	I.	$\left[M^1L^1T^{-2}A^{-2}\right]$
b)	Magnetic field	II.	$\left[L^2A^1\right]$
c)	Magnetic moment	III.	$\left[M^1T^{-2}A^{-1}\right]$
d)	Permeability of free space	IV.	$\left[M^1L^2T^{-2}\right]$

Choose the correct answer from the options given below:

A) $a-III, b-I, c-II, d-IV$ C) $a-IV, b-II, c-III, d-I$

C) $a-IV, b-III, c-II, d-I$ D) $a-I, b-III, c-II, d-IV$

18. Electromagnetic waves travel in a medium with speed of $1.5\times10^8\ ms^{-1}$. The relative permeability of the medium is 2.0. The relative permittivity will be:

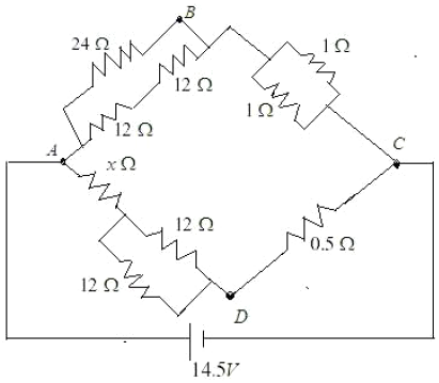
A) 1 B) 4 C) 5 D) 2

19. The ratio of the shortest wavelength of Balmer series to the shortest wavelength of Lyman series for hydrogen atom is:

A) 2:1 B) 1:4 C) 1:2 D) 4:1

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20. The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is:

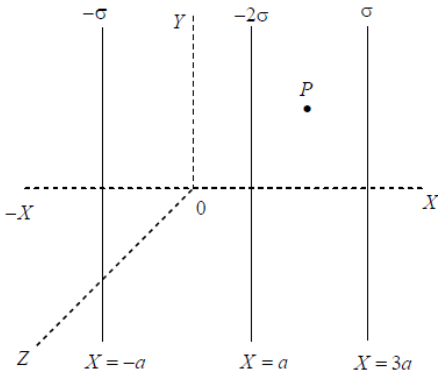


A) 6Ω B) 9Ω C) 42Ω D)3Ω

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Three infinitely long charged thin sheets are placed as shown in figure. The magnitude of electric fields at the point P is $\frac{x\sigma}{\epsilon_0}$. The value of x is _____(all quantities are measured in SI units).



22. A wire of resistance R and radius r is stretched till its radius became r/2. If new resistance of the stretched wire is x R, then Value of x is ____.

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23. If the radius of earth is reduced to three-fourth of its present value without change in its mass then value of duration of the day of earth will be _____ hours 30 minutes.

24. A big drop is formed by coalescing 1000 small droplets of water. The ratio of surface energy of 1000 droplets to that of surface energy of big drop is $\frac{10}{x}$. The value of x is _____.

25. Radius of a certain orbit of hydrogen atom is 8.48 Å. If energy of electron in this orbit is E/x. then x = _____.

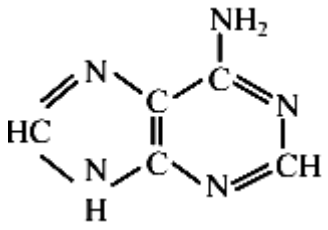
(Given a_0 = 0.53 Å, E = energy of electron in ground state)

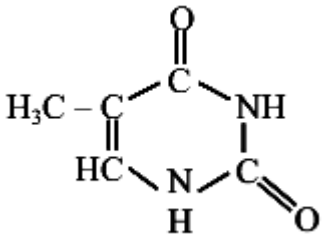
CHEMISTRY **MAX.MARKS: 100**

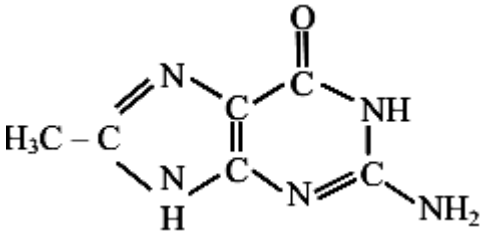
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

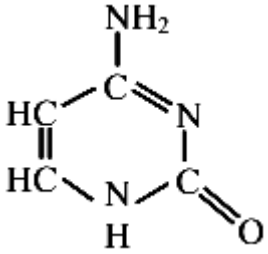
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26. DNA molecule contains 4 bases whose structure are shown below, One of the structures is not correct, identify the incorrect base structure.

A) 

B) 

C) 

D) 

27. The number of element(s) from the following that do not belong to lanthanoids is Eu, Cm, Er, Tb, Yb and Lu

A) 5 B) 4 C) 1 D) 3

28. Which of the following material is not a semiconductor.

A) Germanium B) Graphite C) Copper oxide D) Silicon

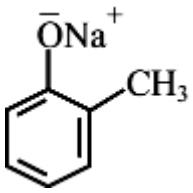
29. The density of 'x' M solution of NaOH is 1.12 g mL^{-1}, while in molality, the concentration of the solution is 3 m. Then *x* is

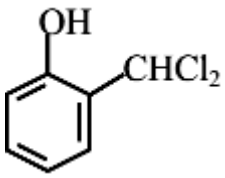
(Given: Molar mass of NaOH is 40 g/mol)

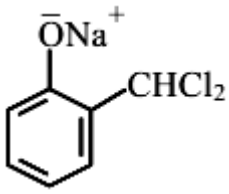
A) 3.5 B) 3.8 C) 2.8 D) 3.0

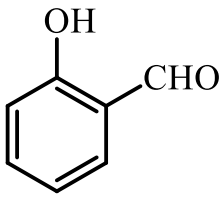
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30. In Reimer –Tiemann reaction, phenol is converted into salicylaldehyde through an intermediate. The structure of intermediate is _____.

A) 

B) 

C) 

D) 

31. Given below are two statements:

Statement I : Picric acid is 2, 4, 6-trinitrotoluene.

Statement II: Phenol-2,4-disulphonic acid is treated with Conc. HNO_3 to get picric acid.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) Both Statement I and Statement II are incorrect

B) Both Statement I and Statement II are correct

C) Statement I is incorrect but Statement II is correct

D) Statement I is correct but Statement II is incorrect

32. Match List I with List II

	List – I (Precipitating reagent and conditions)		**List – III (cation)**
a)	$NH_4Cl + NH_4OH$	I)	Mn^{2+}
b)	$NH_4OH + Na_2CO_3$	II)	Pb^{2+}
c)	$NH_4OH + NH_4Cl + H_2S_{(g)}$	III)	Al^{3+}
d)	$dil.HCl$	IV	Sr^{2+}

Choose the correct answer from the options given below:

A) $a-IV,\ b-III,\ c-I,\ d-II$ B) $a-III,\ b-IV,\ c-I,\ d-II$

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C) $a-IV, b-III, c-II, d-I$ D) $a-III, b-IV, c-II, d-I$

33. Given below are two statements:

Statement I: Gallium is used in the manufacturing of thermometers.

Statement II: A thermometer containing gallium is useful for measuring the freezing point (256 K) of brine solution.

In the light of the above statements, choose the correct answer from the options given below:

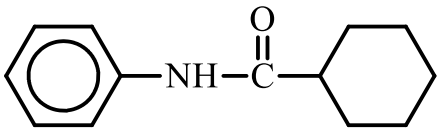
A) Statement I is false but Statement II is true

B) Statement I is true but Statement II is false

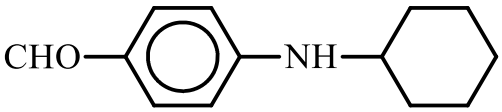
C) Both Statement I and Statement II are false

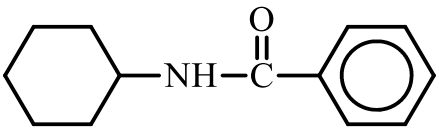
D) Both Statement I and Statement II are true

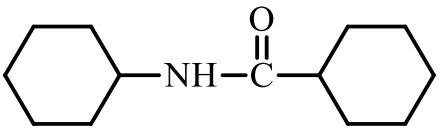
34. Which of the following is metamer of the given compound (X)?

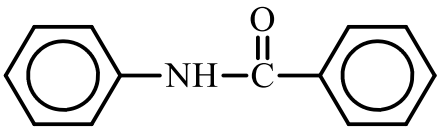


(X)

A) 

B) 

C) 

D) 

35. Functional group present in sulphonic acids is:

A) SO_3H B) SO_2 C) SO_4H D) $-\overset{\overset{O}{\|}}{S}-OH$

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36. Consider the dissociation of the weak acid HX as given below

$HX(aq) \rightleftharpoons H^+(aq) + X^-(aq)$, $Ka = 1.2 \times 10^{-5}$

[K_a : dissociation constant]. The osmotic pressure of 0.03 M aqueous solution of HX at 300 K is $x \times 10^{-2}$ bar. The value of x will be

A) 76.194 B) 84.531 C) 66.472 D) 64.213

37. Number of oxygen atoms present in chemical formula of fuming sulphuric acid is

A) 6 B) 4 C) 7 D) 8

38. Which among the following aldehydes is most reactive towards nucleophilic addition reactions?

A) $C_3H_7-\overset{O}{\overset{\|}{C}}-H$ B) $C_2H_5-\overset{O}{\overset{\|}{C}}-H$ C) $H-\overset{O}{\overset{\|}{C}}-H$ D) $CH_3-\overset{O}{\overset{\|}{C}}-H$

39. At – 20° C and 1 atm pressure, a cylinder contains equal number of H_2, I_2 and HI molecules for the reaction at equilibrium

$H_2(g) + I_2(g) \rightleftharpoons 2HI\ g$, the K_p for the process is $x \times 10^{-1}$.

x = _____. [Given: R = 0.082L atm K^{-1} mol^{-1}]

A) 2 B) 0.01 C) 1 D) 10

40. Match List I with List II

LIST-I (Molecule / Species)	**List II (Property / Shape)**
a. SO_2Cl_2	I. Paramagnetic
b. NO	II. Diamagnetic
c. NO_2^-	III. Tetrahedral
d. I_3^-	IV. Linear

Choose the correct answer from the options given below:

A) $a-III, b-IV, c-II, d-I$ B) $a-IV, b-I, c-III, d-II$

C) $a-III, b-I, c-II, d-IV$ D) $a-II, b-III, c-I, d-IV$

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41. Match List I with List II

	List-I (Hybridization)		**List II (Orientation in space**
a.	sp^3	I.	Trigonal bipyramidal
b.	dsp^2	II.	Octahedral
c.	sp^3d	III.	Tetrahedral
d.	sp^3d^2	IV.	Square planar

Choose the correct answer from the options given below:

A) $a\ -\ III,\ b-\ I,\ c-IV,\ d-II$ B) $a-III,\ b-IV,\ c-I,\ d-II$

C) $a-II, b-I, c-IV, d-III$ D) $a-IV, b-III, c-I, d-\ II$

42. Match List I with List II.

	List-I (Compound / Species)		**List II (Shape/Geometry)**
a.	SF_4	I.	Tetrahedral
b.	BrF_3	II.	Pyramidal
c.	BrO_3^-	III.	See saw
d.	NH_4^+	IV.	Bent T-shape

Choose the correct answer from the options given below:

A) $a-II, b-\ IV, c-III, d-I$ B) $a-III, b-II, c-IV, d-I$

C) $a-III, b-IV, c-II, d-I$ D) $a-II, b-III, c-I, d-IV$

43. Match List I with List II.

	List-I (Compound)		**List II (Uses)**
a.	Iodoform	I.	Fire extinguisher
b.	Carbon tetrachloride	II.	Insecticide
c.	CFC	III.	Antiseptic
d.	DDT	IV.	Frefrigerants

Choose the correct answer from the options given below

A) $a-III,\ b-II,\ c-IV,\ d-I$ B) $a-I,\ b-II, c-III, d-IV$

C) A – II, B – IV, C – I, D – III D) $a-III, b-I,\ c-IV, d-II$

44. Consider the following complexes

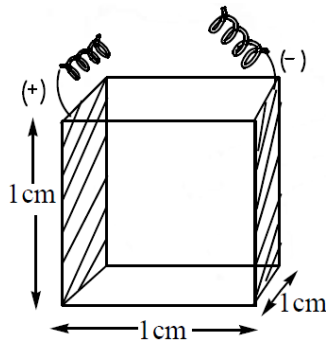
$\left[CoCl(NH_3)_5\right]^{2+}$, $\left[Co(CN)_6\right]^{3-}$, $\left[Co(NH_3)_5(H_2O)\right]^{3+}$, $\left[Cu(H_2O)_4\right]^{2+}$

(a) (b) (c) (d)

The correct order of A, B, C and D in terms of *wavenumber* of light absorbed is:

A) $c<d<a<b$ B) $b<c<a<d$ C) $a<c<b<d$ D) $d<a<c<b$

45. A conductivity cell with two electrodes (dark side) are half filled with infinitely dilute aqueous solution of a weak electrolyte. If volume is doubled by adding more water at constant temperature, the molar conductivity of the cell will –



A) decrease sharply

B) remain same or cannot be measured accurately

C) depend upon type of electrolyte

D) increase sharply

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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46. 9.3 g of pure aniline upon diazotization followed by coupling with phenol gives an orange dye. The mass of orange dye produced (assume 100% yield/conversion) is _____g. (nearest integer)

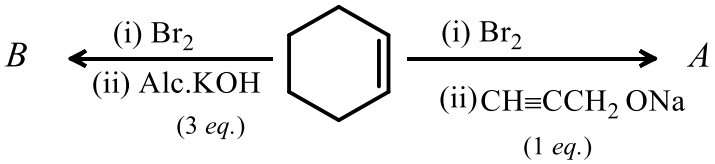
47. The major product of the following reaction is P.

$$CH_3 - C \equiv C - CH_3 \xrightarrow[273K]{\substack{(i)\ Na/liq.NH_3 \\ (ii)\ dil.KMnO_4}} 'p'$$

Number of oxygen atoms present in product 'P' is _____

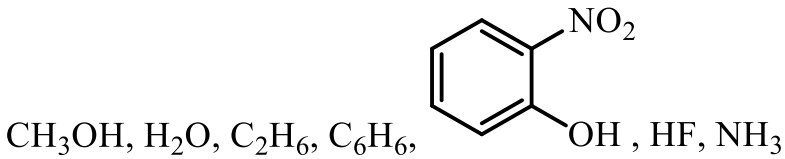
48. Time required for 99.9% completion of a first order reaction is ____ times the time required for completion of 90% reaction.

49. The major products from the following reaction sequence are product A and product B.



The total sum of π electrons in product 'A' and product 'B' are ____

50. Number of molecules from the following which can exhibit hydrogen bonding is _____.



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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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51. The mean and standard deviation of 20 observations are found to be 10 and 2, respectively. On rechecking, it was found that an observation by mistake was taken 8 instead of 12. The correct standard deviation is

A) 1.8 B) $\sqrt{3.86}$ C) $\sqrt{3.96}$ D) 1.94

52. A circle is inscribed in an equilateral triangle of side of length 12. If the area and perimeter of any square inscribed in this circle are m and n, respectively, then $m + n^2$ is equal to

A) 312 B) 396 C) 414 D) 408

53. $\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{\left(\cos^3 x + \sin^3 x\right)^2} dx$ is equal to

A) 1/12 B) 1/9 C) 1/3 D) 1/6

54. A company has two plants A and B to manufacture motorcycles. 60% motorcycles are manufactured at plant A and the remaining are manufactured at plant B. 80% of the motorcycles manufactured at plant A are rated of the standard quality, while 90% of the motorcycles manufactured at plant B are rated of the standard quality. A motorcycle picked up randomly from the total production is found to be of the standard quality. If p is the probability that it was manufactured at plant B, then 126p is

A) 54 B) 66 C) 64 D) 56

55. If $A(3,1,-1)$, $B\left(\frac{5}{3},\frac{7}{3},\frac{1}{3}\right)$, $C(2,2,1)$, and $D\left(\frac{10}{3},\frac{2}{3},\frac{-1}{3}\right)$ are the vertices of a quadrilateral ABCD, then its area is

A) $\frac{2\sqrt{2}}{3}$ B) $2\sqrt{2}$ C) $\frac{5\sqrt{2}}{3}$ D) $\frac{4\sqrt{2}}{3}$

56. Let a variable line of slope $m > 0$ passing through the point (4, –9) intersect the coordinate axes at the points A and B. The minimum value of the sum of the distances of A and B from the origin is

A) 25 B) 30 C) 15 D) 10

57. The number of triangles whose vertices are at the vertices of a regular octagon but none of whose sides is a side of the octagon is

A) 16 B) 24 C) 56 D) 48

58. For $\alpha,\beta\in R$ and a natural number n, Let $A_r=\begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$. Then $2A_{10}-A_8$ is

A) $2\alpha+4\beta$ B) $4\alpha+2\beta$ C) $2n$ D) 0

59. The shortest distance between the lines $\frac{x-3}{2}=\frac{y+15}{-7}=\frac{z-9}{5}$ and $\frac{x+1}{2}=\frac{y-1}{1}=\frac{z-9}{-3}$ is

A) $8\sqrt{3}$ B) $4\sqrt{3}$ C) $5\sqrt{3}$ D) $6\sqrt{3}$

60. For $n\in N$; if $\cot^{-1}3+\cot^{-1}4+\cot^{-1}5+\cot^{-1}n=\frac{\pi}{4}$, then n is equal to _____.

A) 46 B) 43 C) 47 D) 48

61. The interval in which the function $f(x)=x^x, x>0$, is strictly increasing is

A) $\left[\frac{1}{e^2},1\right)$ B) $\left[\frac{1}{e},\infty\right)$ C) $(0,\infty)$ D) $\left(0,\frac{1}{e}\right]$

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62. Let $y = y(x)$ be the solution of the differential equation $\left(1+x^2\right)\frac{dy}{dx}+y=e^{\tan^{-1}x}$, $y(1)=0$. Then $y(0)$ is

A) $\frac{1}{2}\left(e^{\pi/2}-1\right)$ B) $\frac{1}{4}\left(e^{\pi/2}-1\right)$ C) $\frac{1}{4}\left(1-e^{\pi/2}\right)$ D) $\frac{1}{2}\left(1-e^{\pi/2}\right)$

63. The function $f: R \to R$ defined by $f(x)=\frac{x^2+2x-15}{x^2-4x+9}$ is

A) onto but not one-one B) both one-one and onto

C) neither one-one nor onto D) one-one but not onto

64. Let α, β be the distinct roots of the equation $x^2-\left(t^2-5t+6\right)x+1=0,\ t \in R$ and $a_n=\alpha^n+\beta^n$. Then the minimum value of $\frac{a_{2023}+a_{2025}}{a_{2024}}$ is

A) 1/4 B) 1/4 C) -1/2 D) -1/4

65. Let the relations R_1 and R_2 on the set $X=\{1,2,3,.....,20\}$ be given by $R_1=\{(x,y): 2x-3y=2\}$ and $R_2=\{(x,y): -5x+4y=0\}$. If M and N be the minimum number of elements required to be added in R_1 and R_2 respectively, in order to make the relations symmetric, then $M + N$ equals

A) 8 B)16 C) 12 D) 10

66. Let C be the circle of minimum area touching the parabola $y = 6 – x^2$ and the lines $y=\sqrt{3}|x|$. Then, which one of the following points lies on the circle C?

A) (2, 4) B) (1, 1) C) (1, 2) D) (2, 2)

67. Let $y = y(x)$ be the solution of the differential equation $\left(2x\log_e x\right)\frac{dy}{dx}+2y=\frac{3}{x}\log_e x, x>0$ and $y\left(e^{-1}\right)=0$. Then, y(e) is equal to

A) $-\frac{3}{e}$ B) $-\frac{2}{e}$ C) $-\frac{2}{3e}$ D) $-\frac{3}{2e}$

68. Let A = { $n \in [100,700] \cap N : n$ is neither a multiple of 3 nor a multiple of 4}. Then the number of elements in A is

A) 300 B) 310 C) 290 D) 280

69. Let the area of the region enclosed by the curves $y = 3x$, $2y = 27 - 3x$ and $y = 3x - x\sqrt{x}$ be A. Then 10 A is equal to

A) 154 B) 172 C) 162 D) 184

70. If $f(x) = \begin{cases} x^3 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, then

A) $f''\left(\frac{2}{\pi}\right) = \frac{24 - \pi^2}{2\pi}$ B) $f''(0) = 0$

C) $f''(0) = 1$ D) $f''\left(\frac{2}{\pi}\right) = \frac{12 - \pi^2}{2\pi}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $\alpha\beta\gamma = 45;\ \alpha, \beta, \gamma \in R$. If $x(\alpha,1,2) + y(1,\beta,2) + z(2,3,\gamma) = (0,0,0)$ for some $x, y, z \in R$, $xyz \neq 0$, then $6\alpha + 4\beta + \gamma$ is equal to _____

72. Let P be the point (10, –2, –1) and Q be the foot of the perpendicular drawn from the point $R(1,7,6)$ on the line passing through the points $(2,-5,11)$ and $(-6,7,-5)$. Then the length of the line segment PQ is equal to _____

73. Let a conic C pass through the point (4, –2) and P(x, y), $x \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and (3, –5). If the focal distance of the point (7, 1) on C is d, then 12d equals _____

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74. Let x_1, x_2, x_3, x_4 be the solution of the equation $4x^4+8x^3-17x^2-12x+9=0$ and $\left(4+x_1^2\right)\left(4+x_2^2\right)\left(4+x_3^2\right)\left(4+x_4^2\right)=\frac{125}{16}m$. Then the value of m is _____

75. If the second, third and fourth term in the expansion of $(x+y)^n$ are 135, 30 and $\frac{10}{3}$, respectively, then $6\left(n^3+x^2+y\right)$ is equal to _____

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